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Structuring a scientific publication—what authors should pay attention to

Elmar Herbst¹ · Sebastian Kopf² · Daniel Günther³ · AGA Research Committee¹ Department of Trauma, Hand and Reconstructive Surgery, University Hospital Münster, Münster, Germany² Center of Orthopaedics and Traumatology, University Hospital Brandenburg an der Havel, Medical School Brandenburg, Brandenburg, Germany³ Department of Orthopedic Surgery, Trauma Surgery, and Sports Medicine, Cologne Merheim Medical Center, Witten/Herdecke University, Witten, Germany**Abstract**

This article gives a brief overview on how to prepare a scientific publication. Considerations relevant before preparing the manuscript are presented (i.e., preparation of a bulleted outline), and tips are provided for choosing an appropriate and catchy title. When preparing a manuscript, the introduction section is often too long. Here, authors should directly present the scientific/clinical problem, current knowledge on the topic, and the research question, followed by a clear hypothesis. The methods section should be prepared as a detailed recipe in order to make the study reproducible for others. The results section should focus on the primary and secondary endpoints of the study, with key results highlighted in figures and tables. In the discussion section, these endpoints should be discussed on the basis of current literature after a short summary of the main findings. Importantly, results should be discussed on the basis of scientific and clinical relevance rather than in terms of statistical significance.

Keywords

Research · Science · Publication bias · Journal article · Literature

Introduction

The final and sometimes most difficult step in a scientific project is the drafting of a publication. The aim of publication is not only to potentially raise one's scientific profile, but also to facilitate the exchange of research results with colleagues. Although for many people this final part of a scientific project is the most challenging, not publishing results leads to so-called publication bias and is therefore not in line with good scientific practice. Furthermore, failing to complete a scientific project due to a lack of publication would be a waste of the extensive preparatory work that has been done.

In general, manuscript preparation requires familiarity with the current literature relevant to the topic. It is assumed that the

authors have already carried out extensive literature searches as part of the preparation and implementation of the project. Nevertheless, the authors should review the current literature before writing the manuscript to keep track of latest updates on the topic of interest. Subsequently, the authors should review the study hypothesis and the key message(s) to be conveyed to the readers before drafting the manuscript.

Linguistically, scientific publications adhere to concise and straightforward sentences, thus avoiding complexity. In addition, simple and clear language should be used, especially considering that many readers may not be native speakers. Each paragraph should be able to stand alone, consisting of a short introduction, a main body, and a conclusion.



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With few exceptions (e.g., review articles), a scientific publication follows the IMRAD scheme after the title and abstract [5], which consists of:

- introduction,
- materials and methods,
- results, and
- discussion;
- often, a conclusion follows the discussion section.

Before starting to write the manuscript, authors should familiarize themselves with the specific requirements of the target journal in order to adhere to the correct structural and formal criteria (e.g., abstract structure, word count, number of figures, citation format, etc.). This saves time-consuming revisions after publication. Since the manuscript should follow a clear structure, it is advisable to create a short outline with bullet points before writing individual sections. This helps to focus on the essential content of the publication and avoid unnecessarily long manuscripts. In this context, the author needs to consider what messages he or she wants to convey to the reader.

After the first draft of the manuscript has been written, the paper must be submitted to all coauthors and undergo an internal review process. Not only must all authors have full access to the manuscript, but all coauthors must be familiar with its content. Furthermore, the internal review process has the advantage that the manuscript is reviewed and corrected by different people. This is a very valuable experience, especially for inexperienced authors.

Structuring an article

Title selection

The title of the publication should ideally be free of abbreviations and not pose a question. It is important that the title be concise and clear, allowing the reader to quickly infer the content of the publication. Some academic journals now require the key message of the publication to be included in the title.

Titles such as “20-year follow-up after medial unicondylar hemiprosthesis” are relatively meaningless, whereas “High

implant survival and patient satisfaction 20 years after medial unicondylar hemiprosthesis at the knee joint” gives a good idea of the content of the publication.

It is important that special study designs (e.g., systematic reviews, meta-analyses, and prospective randomized trials) are identified as such in the title.

The abstract should follow immediately after the title. How to write scientific abstracts is explained in a separate article.

Introduction

The introduction, as the name suggests, should briefly introduce the topic and engage the reader in such a way that they want to read the rest of the manuscript. Authors should address the following key questions [2, 5]:

- Why is the study being carried out or what problem is being addressed?
- What is known about the topic and what are the open questions?

Essentially, the introduction serves to answer these core questions. However, authors often spend a lot of energy listing very general background information. For example, the information that nearly 130,000 ACL tears occur each year in the United States alone is interesting, but irrelevant to most scientific papers, especially those with a biomechanical study design [9]. Nevertheless, such sentences, sometimes containing common knowledge, appear regularly, especially in the first paragraphs. It is recommended that authors emphasize the clinical or scientific problem from the beginning. For example, introductory sentences such as “A commonly observed problems with valgus-producing high tibial osteotomies are hinge fractures” are appropriate [6]. With such statements, the reader is immediately engaged and recognizes from the first sentence that the paper will address the problem of hinge fractures.

Another common mistake is to present in detail controversies in the current literature. This belongs in the discussion section and should not be included in the introduction. Instead, the authors should indicate that the literature might not be conclusive, but should not discuss the strengths and

weaknesses and reasons for discrepant results of other studies [2].

After providing the necessary background and rationale for conducting the study, authors should state a clear study aim or research question in a short paragraph. A good research question is one that can be answered with a clear “yes” or “no,” or with a clear “outcome.” Statements such as “The aim of this work was to investigate long-term outcomes after medial unicondylar hemiprosthesis” are not very informative. Instead, the aim of the study could be worded as follows: “The aim of the study was to investigate how many unicondylar hemiprostheses had to be converted to total arthroplasty 20 years after implantation.”

After formulating the research question, a scientifically sound hypothesis for the study must be written. The hypothesis must also be understandable and as specific as possible. Phrases such as “It was hypothesized that interference screw fixation is superior to extracortical fixation systems” are not recommended, as they are too broad. Alternatively, a hypothesis could be as follows: “It was hypothesized that the maximum failure load in Newtons and stiffness in N/mm after interference screw fixation would be significantly higher than after extracortical graft fixation.”

As the clinical/scientific problem and the resultant study idea including research question and hypotheses are established before conducting the study, it is recommended to write the introduction section at a very early stage, as relevant text passages for the introduction may already have been written for ethics and/or grant applications, etc.

Materials and methods

In this section, the study population, study procedure, primary and secondary endpoints, and data management are described in detail, like a recipe. The methods should be described in as much detail as possible so that they can be understood and replicated by other scientists. To be as efficient as possible in writing the methodology, it is recommended to write this section while conducting the study. If parts of the methodology or techniques

have been adopted from previous work, this part can be shortened and the authors can simply refer to these previous studies [2, 5].

The methodology usually starts with a description of the study design (e.g., prospective, randomized, retrospective, biomechanical study on human cadaveric joints, etc.). Phrases such as "retrospective analysis of prospectively collected data" should be avoided, as this is ultimately a retrospective study and should be clearly identified as such. In addition, the necessary information about the consultation and approval of the responsible ethics committee must be provided.

Next, the study population must be described in detail, starting with the predefined inclusion and exclusion criteria, including the justification for the chosen criteria. For clinical trials, the study period should also be defined. Ideally, for clinical trials, the number of patients who were eligible for the study should be noted, as well as those who were ultimately included and excluded, including the reasons for exclusion from the trial. In addition, it should be noted how many of the included patients were followed up and how many could not be followed up. In the current literature, it is accepted that the rate of patients lost to follow-up should be less than 20% (the so-called loss to follow-up rate). Ideally, all these data should be presented in a flowchart. For randomized trials, the CONSORT (Consolidated Standards of Reporting Trials) checklist is recommended [10], which can also be used for all other clinical trials. Only a precise description of the study population can prevent criticism of selection bias, i.e., the distortion of the results due to (too) selective inclusion of patients [11].

After describing the study cohort, the scientific procedures are described in detail. For experimental work, it is advisable to use schematic drawings of the experimental setups or photographs with a neutral background to better illustrate the methods.

When explaining the methodology or experimental setup, it is often forgotten to mention how accurate the measurement method is, or whether or not the methods have been validated before. This is very important for the interpretation of

the data. Similarly, inter- and intra-rater reliability tests should be carried out for measurements that are subject to a degree of subjectivity. This means that the measurements should be repeated by the examiner at a defined interval, and also by an independent person for quality control, in order to analyze variations within an examiner or between two or more raters [2].

For clinical trials that use randomization or matching, where each participant in one group is matched to a patient in the other group on the basis of various factors (e.g., age, sex, body mass index), the randomization and/or matching process should be described in detail.

After describing the scientific workflow, the primary and secondary endpoints of the study (e.g., biomechanical parameters, clinical scores) should be explained based on the previously formulated hypothesis. It must be ensured that the relevant endpoints have been validated for the topic under investigation. This is especially true for new procedures or clinical trials. With regard to clinical endpoints, there are already various recommendations regarding objective and subjective outcome scores [1, 8, 13].

The methodology should conclude with a description of how the data were processed or analyzed. It is essential that the data have been checked for normal distribution and that the statistical tests (parametric and non-parametric) and data presentation (e.g., mean versus median) have been chosen accordingly. In addition, a sample size calculation should have been performed a priori (i.e., before the start of the study). In this regard, the authors need to explain what the basis of the sample size calculation was, how the effect size was calculated, and what power (usually > 80%) was calculated. To avoid problems associated with unexpected dropouts, about 10% more participants per group should be included in the study than the sample size calculation suggests [2].

A common mistake in the presentation of the methodology is that parts of the results are already presented in the methodology. Ultimately, the only purpose of the methodology is to describe the study in a reproducible way.

Results

The presentation of the results also follows a clear structure, with an initial description of the study cohort and demographics. It is recommended to keep the text as short as possible and to summarize demographic information in a table. In general, authors should try to keep absolute numbers in the main text to a minimum and present these quantitative results in tables or figures. A common mistake in the results section is the use of reinforcing adjectives (e.g., "enormous," "pronounced," "clear," etc.), which should be avoided. Another mistake is the (subjective) interpretation of the results, which should be reserved for the discussion. Depending on the measurement accuracy, absolute values should be limited to one decimal only, as smaller numerical values usually have no clinical value and may also be outside the measurement accuracy [2].

After the description of the study cohort, the results are presented according to the primary and secondary endpoints. Random ancillary analyses or findings should only be presented if they add significant value. Authors should focus primarily on answering the hypothesis.

Results are often presented both in tables and figures and in the text. However, such repetition should be avoided. Important quantitative results are best presented in figures (e.g., bar charts), while study cohorts or relevant numerical data can be listed in tables. It is important to ensure that tables and figures are self-explanatory. This means, especially for figures, that the figure legend or caption should provide a good summary of the figure [2].

When figures are used, meaningful and clear axis labels should be used. Figures should be kept as simple as possible so that they can be understood immediately by the reader [12]. Consistent axis scaling should also be used. Unnecessary information or details (e.g., grid lines) and three-dimensional bars should be avoided (■ Fig. 1). If study results are statistically significant, this should be noted.

Bar graphs are the most used bar diagrams as they are clear and easy to interpret (■ Fig. 2); for non-normally distributed data, boxplots have become established. Boxplots indicate the minimum

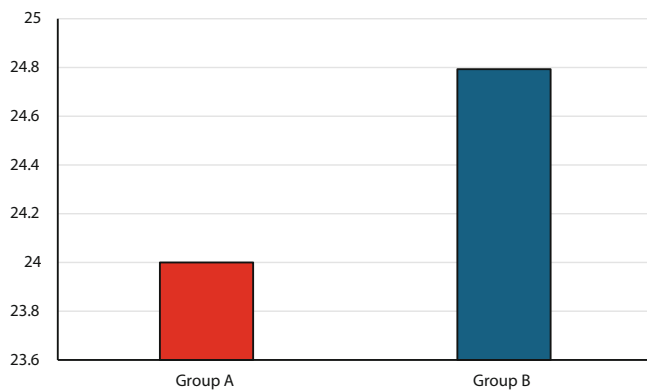


Fig. 1 ▲ Three-dimensional bar graph without error bars. In the y-axis decimals are presented and the *grid lines* are not informative. In general, such a bar graph is difficult to read. Thus, authors should avoid any unnecessary element or information in figures

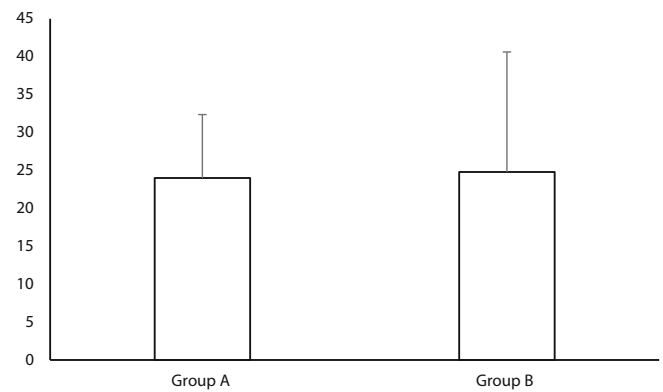


Fig. 2 ▲ Here, two fictional datasets are presented based on their mean and standard deviation. It is clearly visible that both groups have comparable means, but group B has a larger standard deviation. As mentioned earlier, figures must be accompanied by correct axis labels, including units

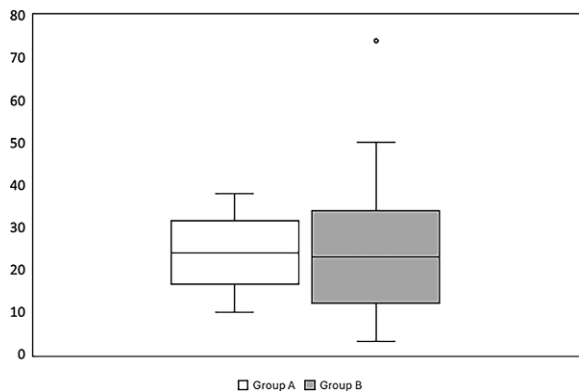


Fig. 3 ▲ This is the box plot of the two groups from **Fig. 2**. In contrast to the bar graph in **Fig. 2**, a boxplot contains significantly more information, particularly regarding the distribution of the data. The median of both groups (represented by the *horizontal line* inside the *box*) is comparable. However, group B exhibits a significantly larger data distribution (the *lower end of the box* represents the lower quartile, while the *upper end* shows the upper or 75th quartile). Additionally, the *outlier* is well presented by the point at 74 in group B

and maximum values, the lower and upper quartile, and the median (**Fig. 3**). Diagrams other than boxplots or bar graphs should be avoided whenever possible [2].

Discussion and conclusion

The aim of the discussion is to discuss one's own results in relation to the current literature, in particular with regard to their clinical and/or scientific relevance. A review of the entire literature is often included in the discussion. However, this unnecessarily lengthens the manuscript and relativizes the significance of the own results. Phrases such as "To the best of our knowledge, this is the first study ..." or "Future studies are needed to ..." should

be avoided, as they convey little content [2].

The discussion should not discuss or describe anything that was not directly investigated in the present study. It is important to maintain a consistent thread that has already been outlined in the introduction and before writing the manuscript. For this purpose, it is advisable to make a bulleted list of the points to be discussed and to stick strictly to this list.

At the beginning of the discussion, the main findings should be summarized. Phrases such as "The main finding of this study was ..." are recommended. However, the results should not be repeated in full. Instead, the hypothesis should be answered.

Then, starting from the primary endpoints, the current literature should be summarized or discussed in the light of the present results, and any discrepancies should be explained. The author of the discussion must always bear in mind that the reader may not be as familiar with the subject as he or she is. This means that scientifically sound and clinically relevant explanations should be provided, while speculation and assumptions should be avoided. When discussing clinical trial results with subjective outcome scores, the focus is not only on discussing statistical significance, but also on discussing clinical relevance. In this context, the so-called minimal clinically important difference (MCID), i.e., the smallest change in value perceived by a patient, can be used [7]. If results fall below the MCID thresholds, they may be statistically significant but not clinically relevant.

After discussing the primary endpoints, the secondary endpoints and any relevant adverse events and complications or other important findings outside the primary and secondary endpoints should/can be discussed.

Then, in brief, the limitations of the study should be presented. The authors should address the weaknesses of the study (e.g., sample size, loss to follow-up, study of donors with reduced bone quality, etc.). Sometimes it is advisable to discuss the limitations if comparable studies have been published previously. For example, the use of porcine bone to test implants for cruciate ligament fixation

is a limitation because it is not human tissue, but the bone quality in pigs is comparable to that of a young adult [4].

At the end of the discussion section, a paragraph on the clinical relevance or implications for daily practice should follow. A short concluding paragraph should be written before the authors move on to the conclusion. The conclusion will be included in the discussion or as a separate section, depending on the requirements of the journal. The conclusion should summarize the main results in one or two sentences, accompanied by a sentence on the clinical significance [2]. It is important that the conclusion be based on the findings only.

The format of the reference list depends on the requirements of the journal. To ensure consistent citation style and to avoid confusion during manuscript revision, a citation management software (e.g., EndNote®, Clarivate Philadelphia, PA, USA; Citavi®, Swiss Academic Software, Wädenswil, Switzerland; Zotero®, etc.) should be used. Nevertheless, all citations should be double-checked before submission. In addition to the correct format, it is important to ensure that the correct journal abbreviation has been used. Furthermore, the most recent literature should be considered. Especially at the beginning of a scientific career, too many references are presented. In general, between 20 and 45 references are sufficient for most original articles. In addition, original sources should be cited whenever possible, rather than citing review articles. Furthermore, references from comparable studies should not be blindly accepted without checking their content. Although this may seem logical, it has been shown that in the orthopedic literature, on average, 38% of papers were cited incorrectly [3].

Checklist for preparation of a scientific publication

- Before writing the manuscript, authors should familiarize themselves with the requirements of the target journal and define the key messages they want readers to take away.
- It is advisable to prepare a bulleted outline of the manuscript in order to

keep the thread going throughout the writing process.

- The manuscript should be reviewed by all coauthors.
- The introduction of an article should be as short as possible. Questions about the scientific rationale for the study or the clinical problem should be answered.
- At the end of the introduction, a clear research question and hypothesis must be defined.
- The methodology must accurately reflect the study process, like a recipe. A sample size calculation should be included for most studies.
- The results section of a manuscript contains general information about the study cohort, followed by the primary and secondary endpoints of the study. While the text describes the results qualitatively, important absolute results should be highlighted in tables and figures.
- The discussion should begin with a summary of the main results, followed by a discussion of the primary and secondary endpoints of the study and the limitations.
- Finally, the clinical significance or conclusion should be briefly restated.

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Declarations

Conflict of interest. E. Herbst, S. Kopf, D. Günther, and the AGA Research Committee declare that they have no competing interests.

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Strukturierung einer wissenschaftlichen Publikation – worauf Autoren achten sollten

Der vorliegende Beitrag bietet eine kurze Übersicht, wie man eine wissenschaftliche Arbeit verfasst. Dabei wird darauf hingewiesen, dass bereits vor dem eigentlichen Schreiben eine Gedankenskizze erstellt werden soll, um beim Schreiben nicht den roten Faden zu verlieren. Nach Wahl eines prägnanten Titels werden in der Einleitung das klinische bzw. wissenschaftliche Problem und die bisherigen Kenntnisse zum Thema kurz geschildert. Daraus wird die wissenschaftliche Fragestellung abgeleitet, welcher eine klare Hypothese folgt. Die Methodik muss so detailliert wie ein Kochrezept dargestellt sein, damit die Studie auch reproduzierbar ist. Der Fokus des Ergebnisteils muss auf den vorab definierten primären und sekundären Endpunkten der Studie liegen, wichtige Daten werden dabei auch in Abbildungen/Tabellen präsentiert. In der Diskussion werden zunächst die Hauptergebnisse zusammengefasst und anschließend die Endpunkte der Studie diskutiert und unter Berücksichtigung der aktuellen Literatur interpretiert. Wichtig dabei ist, dass nicht (nur) die statistische Signifikanz, sondern vor allem die klinische Relevanz dargestellt wird.

Schlüsselwörter

Forschung · Wissenschaft · Publikationsbias · Fachartikel · Literatur